Novelty & AI Justification

MRI-ReportGenerator (Cervical Spine MRI Analysis)

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Scope. This document states what is *novel* about MRI-ReportGenerator and *justifies* the AI design choices behind it. It deliberately does **not** re-explain the measurement algorithms or the validation numbers — those are documented in the technical-depth material, the full manuscript, the final validation summaries, and the development journal. This document *references* that body of evidence and argues the contribution.

Medical framing (hard rule). The system produces measurements and *findings flagged for physician review*, never diagnoses. The novelty claims below are framed accordingly.

1 The novelty in one sentence

The contribution is **not** a new neural network. It is a *system design*: a cervical-spine MRI pipeline built from **healthy-anchored, disease-agnostic geometric detectors**, validated by a **threshold-crossing / distribution-separation methodology** purpose-built for a domain where **no public dataset pairs cervical MRI with per-case expert measurements**, and surfaced through a **fully cited threshold catalog** that only ever flags findings for physician review. Each of those four italicised choices is a deliberate, defensible departure from the obvious approach, and together they are what a thin "wrapper over a segmentation model" does not have.

2 Why a non-AI baseline exists — and why automation is justified

The non-AI baseline is **manual radiologist measurement**. It is the right thing to improve on for two cited reasons (full numbers and citations are summarized in the positioning material):

- **Slow.** Cervical MRI reporting runs ≈ 2.7 min median / 3.8 min mean of PACS interaction (Forsberg 2017, PMID 27714473), with a further self-reported 5–10 min/case of manual geometry (Zhu 2024, PMID 38269650); no formal time-motion study isolates the geometry step — itself a documented gap.
- **Unreliable on exactly what we automate.** Inter-observer agreement collapses for the structures this pipeline measures: cord AP ICC 0.66 axial (Grochmal 2018, PMID 29913296), cord-compression metrics ICC 0.35–0.56 (Fehlings 2006, PMID 16816769), cervical disc grading on the original Pfirrmann scale $\kappa = 0.265$ (Urbanschitz 2021, PMID 34966859), and Cobb angle ~ 0.88 for a single trained reader but ~ 0.55 across mixed readers (Sevin 2025, PMID 41011045).

A reproducible, deterministic measurement layer is therefore a genuine improvement, not a solution in search of a problem. This grounds the *Application* positioning.

3 Four novelty axes (what is genuinely different)

3.1 Disease-agnostic detectors anchored on healthy norms

Most “AI for spine MRI” work trains a classifier to recognise a *named pathology* (stenosis, herniation, myelopathy) from labelled diseased examples. We do the opposite: we measure **geometry and signal against a healthy reference**, so a canal of 8 mm reads “narrow, flagged for review” *regardless of cause*. The detector is defined by the normal range, not by a disease class. Consequences that are themselves the novelty:

- It generalises beyond the diseases in any training set — there is no disease label to overfit.
- It needs no diseased training cohort, only a healthy reference distribution, which removes the train/test leakage failure mode that plagues small-cohort medical classifiers.
- Where a published cervical-MRI norm does not exist, the threshold is *cohort-derived* and labelled as such (e.g. the Ha/Hp compression z-screen and the disc/VB AP-ratio cut — both stated as cohort-calibrated, not literature values, with the underlying mechanism cited: Machino 2021, PMID 34098133, for disc widening with degeneration).

3.2 A validation methodology designed for “no per-case ground truth”

The standard evaluation — per-case sensitivity/specificity against a radiologist gold standard — is **impossible on public data** for cervical MRI, because no dataset pairs the images with per-case expert measurements (established by repeated dataset searches). Rather than fabricate a weak label, we designed an evaluation that fits the constraint:

1. **Healthy-anchoring (specificity):** confirm healthy anatomy reads inside the normal range on a healthy cohort (Spine-Generic).
2. **Threshold-crossing / distribution separation (sensitivity proxy):** confirm symptomatic anatomy crosses into the abnormal range on a symptomatic cohort (MMCSD, CSM/CSR), via distribution-separation statistics (Mann–Whitney), never per-case accuracy.

Designing and defending a sound evaluation when the textbook one is unavailable is a methods contribution in its own right (paper §13; J14–J26).

3.3 The physical-dimension scanner-immunity insight

A non-obvious finding shaped the whole evaluation: **physical-dimension (mm) metrics are scanner-immune** and validate *across* datasets, while **intensity/ratio metrics are acquisition-sensitive** and must be validated *within* one dataset. This is why the millimetre canal / SAC / cord measurements (G3) separate cleanly cross-dataset (p=0.0001) but disc *signal* does not, and why disc metrics were validated within-MMCSD only. Recognising this distinction — and structuring the validation around it — is novel methodology, not a parameter choice (J16, J23).

3.4 Cited, auditable assessement — never a black box, never a diagnosis

Every assessed finding maps to a *cited* threshold in the G6 catalog, with provenance, modality caveats, and an honest label when a threshold is a documented borrow or a cohort-derived value. The output is a status (within / review / outside) “flagged for physician review,” never a diagnosis. A radiologist can check each number against its source. This auditability is a deliberate departure from end-to-end diagnostic models and is the appropriate design for a clinical application.

4 Honesty about negatives *is* part of the novelty

A credible system says what does *not* work. We report, and do not hide, three negatives that a results-chasing project would have buried:

- **G4 cervical alignment (Cobb) is NOT a disease discriminator.** On a balanced multi-site cohort (26 healthy vs 41 symptomatic) the effect evaporates: $d = 0.28$, AUC 0.57, $p = 0.32$. An earlier $n = 11$ result ($d = 0.76$) was a lordosis-biased small sample. We report G4 as a validated *measurement*, not a screen (J24, J26).
- **Disc signal and posterior bulge do not discriminate** within the symptomatic cohort (AUC 0.50 each, level-controlled); only the disc/VB AP-ratio carries a modest signal (AUC 0.62) (J23).
- **The compression-fracture abnormal arm is unvalidated** because no labelled cervical-MRI compression-fracture dataset exists (documented limitation; VerSe authors note cervical fractures are rare and usually non-osteoporotic, PMC8082364).

Reporting these as negatives — rather than tuning a number until it looks clean — is the same discipline that makes the positive results (G3 stenosis, p=0.0001) trustworthy.

5 AI justification: why this is the right AI architecture

The pretrained-and-compose design (three frozen segmentation engines feeding our measurement and assessement layer) is the defensible choice for a clinical application:

- **Auditability.** Outputs trace to a measurement and a cited threshold, not an opaque model.
- **No overfitting by construction.** Frozen segmenters + healthy-anchored / cited thresholds mean the system cannot have memorised the evaluation cohort.
- **Graceful degradation.** Per-component error contracts mean one failed measurement (e.g. a missing SPINEPS mask) degrades a single method, not the whole report.
- **The differentiator is the clinical layer.** The measurement validity, the cited assessement, and the report — the parts that decide whether a radiologist trusts the output — are exactly the parts we built and validated.

6 Relation to similarly-scoped work

This project shares a high-level title space with other cervical-spine MRI efforts. The differentiation above is stated on its own terms and does not depend on any one comparator: the combination of disease-agnostic healthy-anchored detectors, a no-per-case-GT validation methodology, the scanner-immunity insight, and a fully cited review-only assessement layer is the contribution. A direct, point-by-point head-to-head against a specific comparator project can be appended here once that project's scope and abstract are available; it is not required to establish the novelty, which stands on the four axes above.

7 Summary

- The novelty is a **system**: healthy-anchored disease-agnostic detectors, a validation methodology built for the no-per-case-GT constraint, the physical-dimension scanner-immunity insight, and a cited review-only assessement layer.
- The AI choice (compose frozen, validated segmenters; build and validate the clinical layer) is **justified**, not a shortcut — it gives auditability, no overfitting, and graceful degradation.

- Credibility comes from **reported negatives** (G4 not a discriminator; disc signal/bulge dead; compression-fracture arm unvalidated) as much as from the strong G3 result.
- Full technical depth, methods, numbers, and development narrative are documented in the companion technical materials and the full manuscript.